System Audit Report

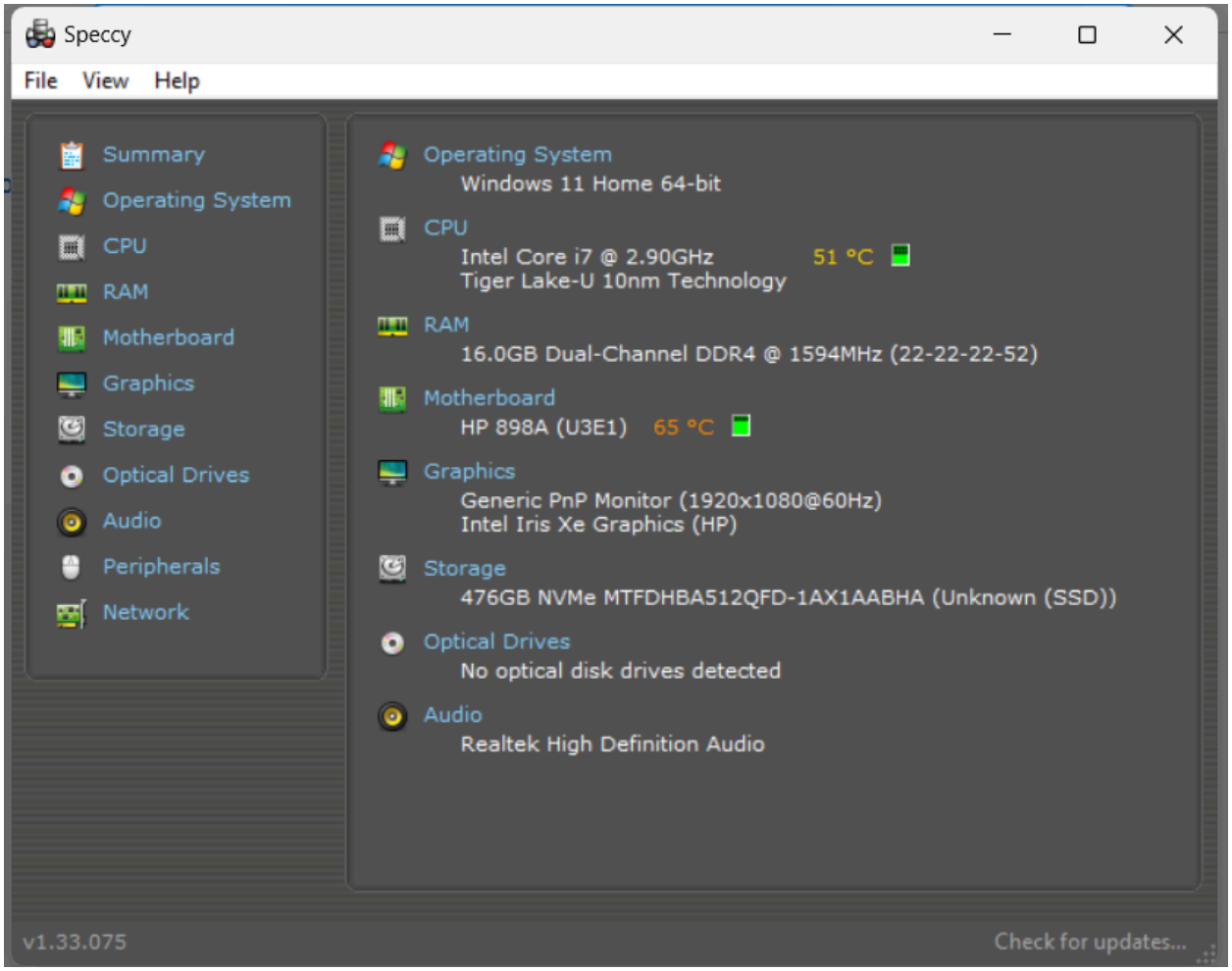
Speccy

1. Executive Summary

This report presents the findings of a comprehensive system audit conducted on the user's personal computer using Speccy v1.33. The audit aimed to gather detailed information regarding the system's hardware components, software environment, and network configuration. Overall, the system demonstrates modern specifications suitable for general computing tasks, with a capable CPU, ample RAM, and fast storage. Several observations regarding storage utilization and network security settings are noted, with recommendations provided for optimization and enhanced security.

2. System Overview (Summary)

Component	Specification
Operating System	Windows 11 Home 64-bit
CPU	Intel Core i7 @ 2.90GHz (Tiger Lake-U 10nm)
RAM	16.0GB Dual-Channel DDR4 @ 1594MHz (22-22-22-52)
Motherboard	HP 8984 (U3E1)
Graphics	Generic PnP Monitor (1920x1080@60Hz), Intel Iris Xe Graphics (HP)
Storage	476GB NVMe MTFDHBA512QFD-1AX1AABHA (SSD)
Optical Drives	No optical disk drives detected
Audio	Realtek High Definition Audio

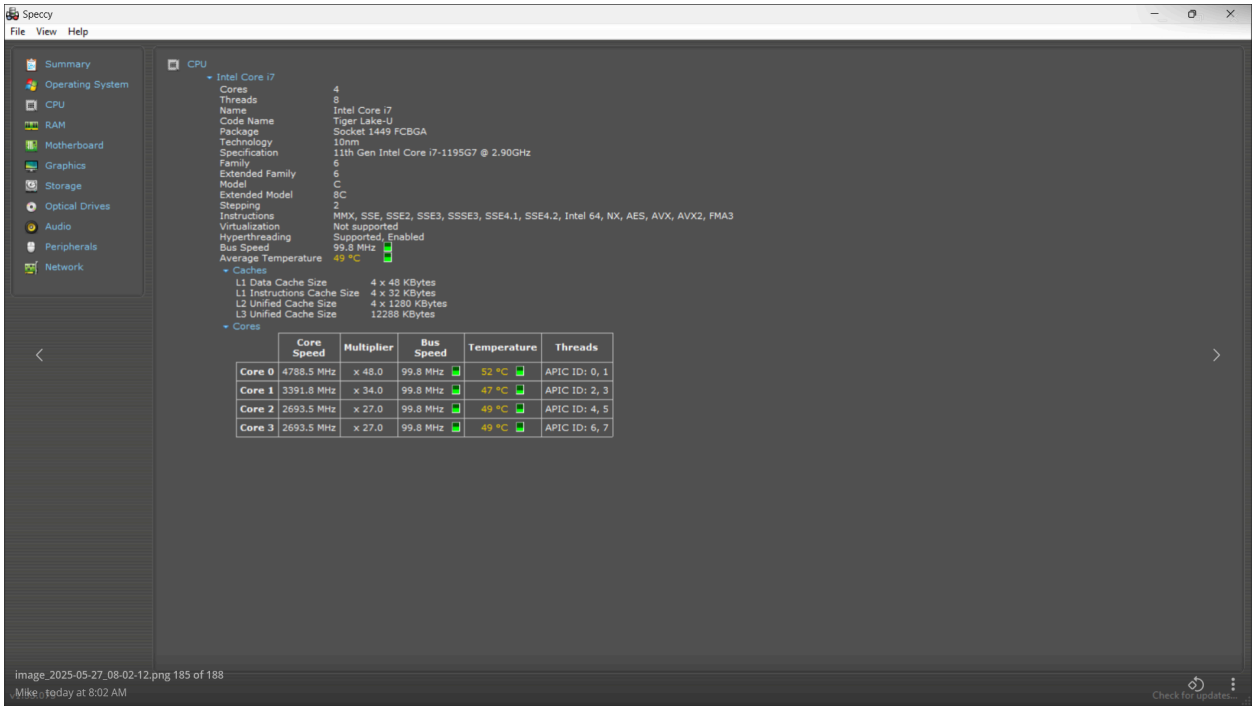


3. Detailed Findings

3.1. Operating System

- **OS Name:** Windows 11 Home 64-bit
- **Observations:** The system is running a current and widely supported operating system, Windows 11 Home (64-bit). This ensures access to the latest features, security updates, and software compatibility.

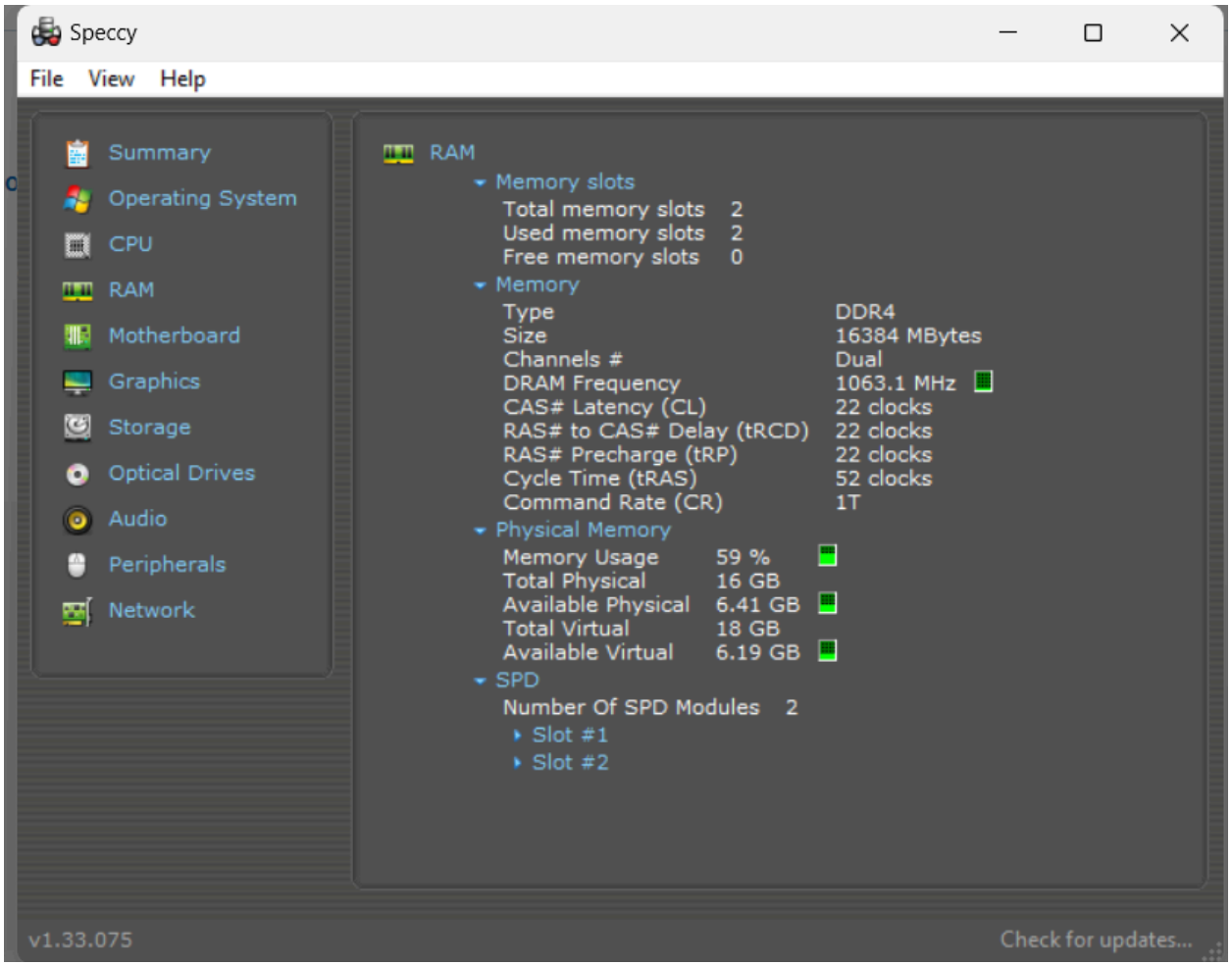
3.2. Central Processing Unit (CPU)



- **Name:** Intel Core i7 @ 2.90GHz
- **Code Name:** Tiger Lake-U
- **Technology:** 10nm
- **Specification:** 11th Gen Intel Core i7-1195G7 @ 2.90GHz
- **Cores/Threads:** 4 Cores / 8 Threads
- **Package:** Socket 1449 FCBGA
- **Instructions:** MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, Intel 64, NX, AES, AVX, AVX2, FMA3
- **Virtualization:** Not supported (Note: This might be a Speccy reporting anomaly or a BIOS setting, as most modern i7s support virtualization.)
- **Hyperthreading:** Supported, Enabled
- **Bus Speed:** 99.8 MHz
- **Average Temperature:** 49°C
- **Cache:**
  - L1 Data Cache Size: 4 x 32 KBytes
  - L1 Instructions Cache Size: 4 x 32 KBytes
  - L2 Unified Cache Size: 4 x 1280 KBytes
  - L3 Unified Cache Size: 12288 KBytes
- **Core Temperatures:**
  - Core 0: 52°C
  - Core 1: 47°C
  - Core 2: 49°C
  - Core 3: 49°C

- **Observations:** The Intel Core i7 (11th Gen) is a capable processor, offering good performance for multitasking, productivity, and moderate content creation. Core temperatures are within acceptable operating limits. The reported "Virtualization: Not supported" should be verified in the BIOS if virtualization features are required.

3.3. Random Access Memory (RAM)



- **Type:** DDR4
- **Size:** 16.0GB (16384 MBytes)
- **Channels:** Dual-Channel
- **DRAM Frequency:** 1063.1 MHz (Effective 2126.2 MHz)
- **CAS# Latency (CL):** 22 clocks
- **RAS# to CAS# Delay (tRCD):** 22 clocks
- **RAS# Precharge (tRP):** 22 clocks
- **Cycle Time (tRAS):** 52 clocks
- **Command Rate (CR):** 1T
- **Memory Slots:** Total: 2, Used: 2, Free: 0

- **Physical Memory Usage:** 59% (6.41 GB Available out of 16 GB Total)
- **Virtual Memory:** Total: 18 GB, Available: 6.19 GB
- **SPD Modules:** 2 modules detected (Slot #1, Slot #2)
- **Observations:** 16GB of DDR4 RAM in a dual-channel configuration provides sufficient memory for most modern applications and multitasking. The current memory usage of 59% indicates a healthy amount of available RAM for typical operations.

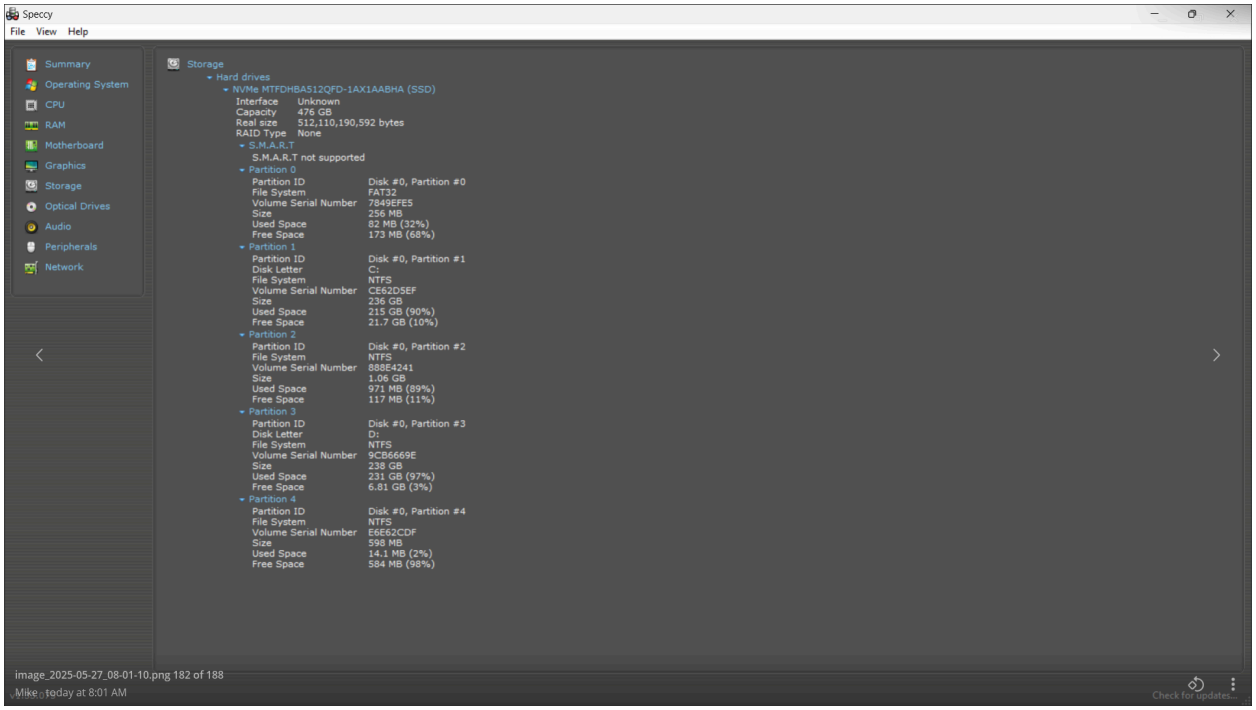
3.4. Motherboard

- **Manufacturer:** HP
- **Model:** 8984 (U3E1)
- **Temperature:** 65°C
- **Observations:** The motherboard temperature is within a normal range for a laptop under typical load. Being an HP proprietary board, specific upgrade options might be limited, but it adequately supports the installed components.

3.5. Graphics

- **Monitor:** Generic PnP Monitor (1920x1080@60Hz)
- **Graphics Adapter:** Intel Iris Xe Graphics (HP)
- **Observations:** The integrated Intel Iris Xe Graphics is suitable for everyday tasks, media consumption, and light gaming. The monitor resolution of 1920x1080 at 60Hz is standard for most laptops, providing a clear visual experience.

3.6. Storage



- **Hard Drive:** 476GB NVMe MTFDHBA512QFD-1AX1AABHA (SSD)
- **Interface:** Unknown (likely NVMe PCIe)
- **Capacity:** 476 GB (512,110,190,592 bytes)
- **RAID Type:** None
- **S.M.A.R.T:** Not supported (Note: This is unusual for an SSD and might indicate a driver issue or a limitation in Speccy's detection for this specific NVMe drive. S.M.A.R.T. is crucial for monitoring drive health.)
- **Partitions:**
  - **Partition 0:** Disk #0, Partition 0
    - File System: FAT32
    - Size: 255 MB
    - Used Space: 82 MB (32%)
    - Free Space: 173 MB (68%)
  - **Partition 1 (C:):** Disk #0, Partition #1
    - File System: NTFS
    - Size: 236 GB
    - Used Space: 215 GB (90%)
    - Free Space: 21.7 GB (10%)
  - **Partition 2:** Disk #0, Partition #2
    - File System: NTFS
    - Size: 1.06 GB
    - Used Space: 971 MB (89%)
    - Free Space: 117 MB (11%)

- **Partition 3 (D:):** Disk #0, Partition #3
  - File System: NTFS
  - Size: 238 GB
  - Used Space: 231 GB (97%)
  - Free Space: 6.81 GB (3%)
- **Partition 4:** Disk #0, Partition #4
  - Size: 598 MB
  - Used Space: 14.1 MB (2%)
  - Free Space: 564 MB (90%)

- **Observations:** The NVMe SSD provides fast storage performance, significantly contributing to overall system responsiveness. However, several partitions (C: and D:) show very high utilization (90% and 97% respectively). This can lead to performance degradation and hinder system updates or application installations. The lack of S.M.A.R.T. support detection for the SSD is a concern and should be investigated.

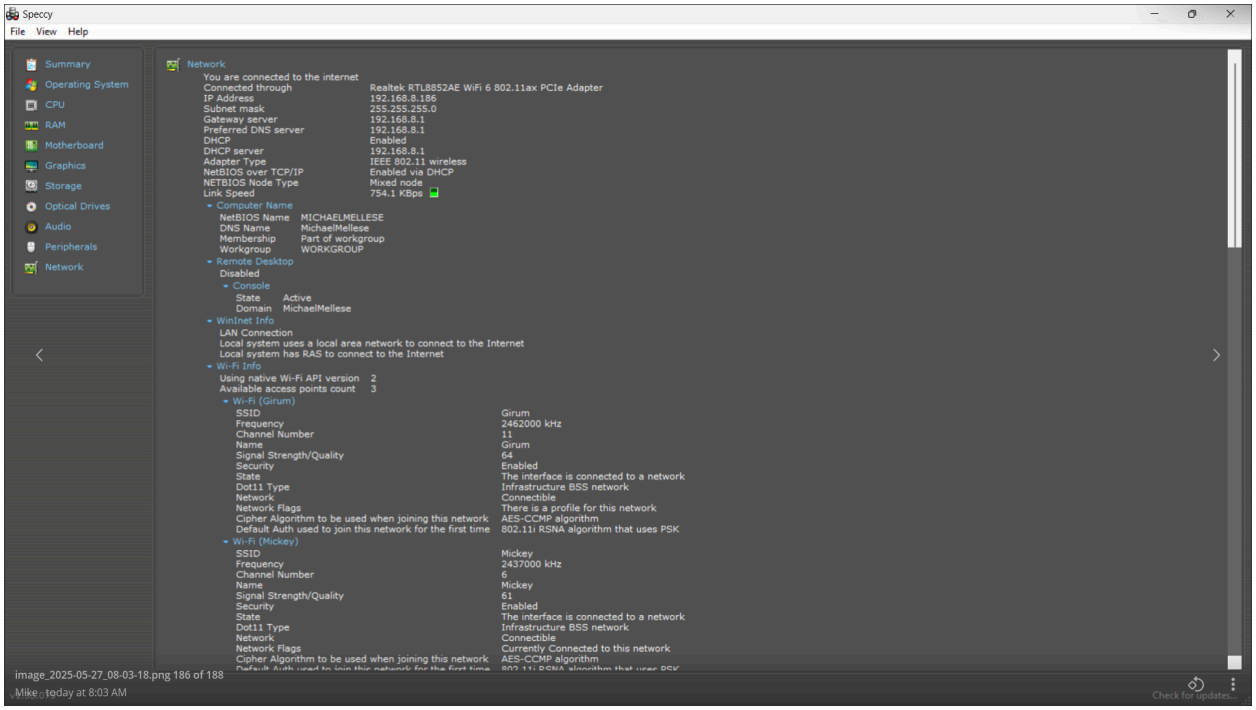
3.7. Optical Drives

- **Status:** No optical disk drives detected.
- **Observations:** This is common for modern laptops, which often omit optical drives in favor of thinner designs and cloud-based media.

3.8. Audio

- **Audio Device:** Realtek High Definition Audio
- **Observations:** Standard integrated audio solution, suitable for general audio playback and communication.

3.9. Network



- **Internet Connection:** Connected
- **Connection Type:** Realtek RTL8852AE WiFi 6 802.11ax PCIe Adapter
- **IP Address:** 192.168.8.186
- **Subnet Mask:** 255.255.255.0
- **Gateway/Preferred DNS Server:** 192.168.8.1
- **DHCP:** Enabled
- **Adapter Type:** IEEE 802.11 wireless
- **Link Speed:** 754.1 KBps (Note: This link speed seems unusually low for Wi-Fi 6. It might be a momentary reading or a reporting error by Speccy.)
- **Computer Name:** MICHAELMELLESE
- **NetBIOS Name:** MICHAELMELLESE
- **DNS Name:** MichaelMellese
- **Membership:** Part of workgroup (WORKGROUP)
- **Remote Desktop:** Disabled
- **Network Discovery:** Disabled
- **File and Printer Sharing:** Disabled
- **Simple File Sharing:** Enabled
- **Administrative Shares:** Enabled
- **Network Access: Sharing and security model for local accounts:** Classic - local users authenticate as themselves.
- **Adapters List:**
  - Bluetooth Device (Personal Area Network)
  - ExpressVPN TAP Adapter (Ethernet 2)

    - ExpressVPN Wintun Driver (Local Area Connection)
    - Realtek RTL8852AE WiFi 6 802.11ax PCIe Adapter (WiFi)
    - TAP-ProtonVPN Windows Adapter V9 (Local Area Connection 2)
    - TAP-Windows Adapter V9 (Local Area Connection 3)
    - VMware Virtual Ethernet Adapter for VNet (VMware Network Adapter VMnet8)
    - VMware Virtual Ethernet Adapter for VNet (VMware Network Adapter VMnet1)
    - VirtualBox Host-Only Ethernet Adapter (Ethernet)
- **Current TCP Connections:** Numerous active connections, including ExpressVPN, Microsoft Edge, McAfee WebAdvisor, qBittorrent, Telegram, uTorrent, svchost.exe, MsMpEng.exe (Microsoft Defender), nessusd.exe, and others.
- **Observations:** The system is connected via Wi-Fi 6, which is a modern standard. The reported link speed is concerningly low and should be re-evaluated. Network Discovery and File and Printer Sharing are disabled, which is good for security in public or untrusted networks. However, "Simple File Sharing" and "Administrative Shares" are enabled, and the security model is "Classic," which authenticates local users as themselves. This combination could pose a security risk if not properly managed, especially if the system is used in a shared environment. The presence of multiple virtual network adapters (ExpressVPN, VMware, VirtualBox) indicates the use of VPN software and virtualization, which is expected. The list of active TCP connections shows a variety of applications, including security software (McAfee, MsMpEng), communication tools (Telegram), and file-sharing clients (qBittorrent, uTorrent).

4. Recommendations

Based on the audit findings, the following recommendations are provided:

1. **Storage Management:**
    - **Action:** Immediately free up space on the C: and D: drives. This can be done by uninstalling unused applications, deleting unnecessary files, moving large files to external storage, or utilizing disk cleanup utilities.
    - **Benefit:** Improves system performance, allows for critical updates, and prevents potential data loss due to insufficient space.
2. **S.M.A.R.T. Status for SSD:**
    - **Action:** Investigate why S.M.A.R.T. is reported as "Not supported" for the NVMe SSD. This may involve updating storage drivers, checking the drive manufacturer's utility, or consulting the system manufacturer's support.
    - **Benefit:** S.M.A.R.T. provides crucial health monitoring for SSDs, allowing proactive measures to be taken before potential drive failure.
3. **Network Link Speed Verification:**

- **Action:** Verify the actual Wi-Fi link speed through Windows Network Status or the router's interface. If the reported 754.1 KBps is accurate, investigate potential causes such as weak signal, interference, or outdated Wi-Fi adapter drivers.
- **Benefit:** Ensures optimal network performance for faster internet access and smoother online activities.

4. **Network Sharing and Security Review:**
   - **Action:** Review the necessity of "Simple File Sharing" and "Administrative Shares" being enabled, especially in conjunction with the "Classic" security model. If not explicitly required for specific tasks, consider disabling them to enhance security. If sharing is needed, ensure strong passwords are in place for all user accounts.
   - **Benefit:** Reduces the attack surface and minimizes the risk of unauthorized access to shared resources.
5. **Software Review:**
   - **Action:** Regularly review installed software, especially file-sharing clients (qBittorrent, uTorrent), to ensure they are up-to-date and configured securely. Be mindful of the content being shared and downloaded to avoid legal or security issues.
   - **Benefit:** Prevents vulnerabilities from outdated software and reduces exposure to malicious content.
6. **Virtualization Support :**
   - **Action:** If virtualization features are desired (e.g., for running virtual machines more efficiently), check the system's BIOS settings to confirm if Intel Virtualization Technology (VT-x) is enabled.
   - **Benefit:** Improves performance and compatibility for virtualization software.

5. Conclusion

The system under audit is a well-equipped Windows 11 machine with a capable Intel Core i7 processor, ample RAM, and fast NVMe storage. While generally performing well, critical attention should be paid to storage utilization to maintain optimal performance and system stability. Furthermore, a review of network sharing settings is recommended to enhance security posture. Addressing these points will contribute to a more efficient, secure, and reliable computing experience.